

Specifications, quantities and costing

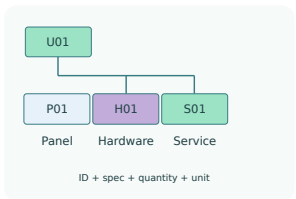
A quantity file links drawings, materials, hardware and services. Build identifiable items with explicit units and a declared training cost model, then check revisions and double counting.

What you will accomplish

- Link part IDs to specifications and quantities
- Calculate each line using matching units
- Check scope and revision before issue

Link part IDs to specifications and quantities

Assign unit U01 and part IDs such as P01. A panel row records material, thickness, dimension stage, face, edges, count and drawing reference. Distinguish a parts list from sheet purchasing: total part area alone does not establish sheet count without layout, grain, kerf and waste review. Separate hardware, finish and services with stated units.



Stable IDs connect drawings, part rows and quantity items.

Calculate each line using matching units

Training prices only: assume 2 sheets at 100 currency units each, 4 hardware sets at 15 per set, and 1 installation service at 40. Line totals are 200, 60 and 40; the total is 300 currency units. These are not market prices. Record quantity × matching unit price, currency, date, source and scope; do not multiply an area rate by a part count.

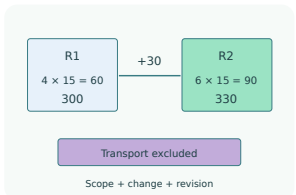
Item	Calc	Total
Sheets	2 × 100	200
Sets	4 × 15	60
Install	1 × 40	40
Sum		300

Training prices; not market rates

Training total: $2 \times 100 + 4 \times 15 + 1 \times 40 = 300$.

Check scope and revision before issue

Define included scope: finish already included in a faced-sheet rate must not be counted again for the same work. List transport, installation, tax and contingency as items or explicit exclusions without inventing universal percentages. Update dependent quantities when a unit changes and compare revisions. Mark an unpriced item as missing information rather than zero-cost work.



Scope checks prevent duplication; missing rates remain visible.

Worked case and decision

The training total was 300 currency units. Increasing hardware sets from 4 to 6 while other inputs stay fixed changes that line from 60 to $6 \times 15 = 90$. The revised total is $200 + 90 + 40 = 330$. Record the 30 increase, revision and the exercise exclusion of transport.

Decision: issue the revised 330 total with the 30 change and exclusions recorded.

Check understanding

What is the line total for 6 sets at 15 per set?

- 90
- 21
- 60

Correct: 90. Quantity $\times$ matching rate: $6 \times 15 = 90$.

Lesson questions

Should an unknown rate be entered as zero?

Mark it as missing; zero can misleadingly imply free or included work.

Practice

Prepare the three training lines in two revisions, with units, assumed prices, scope and cost change.

Quantity and specification table

Cost calculation

Revision change and exclusions

Review your work against the criteria. Saving it does not constitute technical project approval.

- Quantity and rate units match
- Totals are 300 then 330
- No duplicated scope